Dictionary-Based Slang Normalization for Indonesian Sentiment Classification: A Controlled Comparison

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***Abstract*—In this study, dictionary-based slang normalization is evaluated for Indonesian sentiment classification under two uses: training with normalized text and changing inputs to an existing classifier. Word and character TF-IDF support vector machines and IndoBERT are compared on identical audited PRDECT-ID and SmSA splits. Macro-F1, paired bootstrap confidence intervals, direct effect contrasts, and prediction-correctness transitions assess the outcomes. On SmSA, the IndoBERT normalized pipeline gained 0.0202 macro-F1 on average across three seeds, with a nominal 95% confidence interval of [0.0037, 0.0381]. Normalizing inputs to the original-trained model instead yielded a mean change of 0.0006. Fixed-model switches both corrected and introduced errors. Direct contrasts did not establish greater normalization benefits for word-based models than for the alternatives. No setting met the predefined practical-gain criterion requiring the interval's lower bound to exceed 0.01 macro-F1. This study contributes a controlled comparison showing why a separately trained pipeline's benefit alone does not establish an inference-only benefit. The findings support evaluating these two uses separately for each dataset and model, without causally attributing pipeline changes to substitution, retraining, or tuning.**

***Index Terms*—Indonesian sentiment analysis, slang normalization, text preprocessing, IndoBERT, character n-grams**

I. Introduction

Indonesian customer reviews and online comments contain abbreviations, spelling variants, and morphological transformations that differ from standard written forms [1], [2]. Sentiment classifiers can receive this wording directly or replace known informal forms with dictionary alternatives. Substitution may reduce variation, but ambiguous mappings and context-dependent meanings can also change the input inappropriately.

Prior work examines normalization with Indonesian pretrained models [3], [4] and compares broader preprocessing pipelines across representations [5]. This study combines one isolated dictionary intervention, shared audited splits for two public datasets, and separate word and character TF-IDF models alongside IndoBERT. Its contribution is a controlled comparison of two decisions: training with normalized text and applying substitution to an existing classifier. Comparing separately tuned pipelines alongside fixed-model input switches tests whether an observed pipeline benefit also appears at inference, while quantifying corrected and introduced errors.

Three questions guide the study. *RQ1:* Does normalization benefit the word model more than the character model or IndoBERT? *RQ2:* How do the observed normalization effects vary between PRDECT-ID and SmSA? *RQ3:* How does switching between original and normalized inputs affect prediction correctness for a fixed classifier? Direct contrasts address RQ1; dataset-specific effects address RQ2; and counts of corrected and introduced errors in both switch directions address RQ3. Cross-dataset comparisons are descriptive because the datasets have different label schemes and sources. The following review positions this design relative to prior normalization and sentiment studies.

II. Related Work

Indonesian slang normalization addresses the lexical variation found in informal text. The Colloquial Indonesian Lexicon provides informal–formal mappings, including entries with multiple alternatives [1]. Such ambiguity matters because a dictionary substitution does not necessarily preserve the meaning intended in a particular review. IndoCollex evaluates normalization through annotated word transformations [2]; success on that task does not by itself establish improved sentiment classification. Downstream evaluation must therefore assess both the benefits and errors associated with replacing informal forms.

The representation used by the classifier is another relevant factor. IndoNLU supplies SmSA and the IndoBERT pretrained models [6], while IndoBERTweet adapts an Indonesian encoder and vocabulary to Twitter [7]. These resources motivate studying normalization alongside the model's treatment of informal language. Siva et al. [8] compare IndoBERT against an SVM combining word and character TF-IDF. That combined representation leaves their separate responses to normalization unresolved. Word-level lexical matching, character overlap, and contextual subword representations motivate different hypotheses about substitution effects, which require direct empirical comparison.

Recent studies provide direct evidence about preprocessing choices. Bustamin et al. [3] report higher IndoBERT accuracy on tweets after combining dictionary normalization with edit-distance correction; that result concerns the combined intervention. Budianoor et al. [4] isolate slang normalization relative to minimal cleaning on culinary comments, increasing macro-F1 from 0.8272 to 0.8482. Prichanti et al. [5] find different preferences for heavy and minimal preprocessing in Word2Vec/logistic regression and IndoRoBERTa. Their acknowledged lack of component-wise ablation prevents attributing those differences specifically to slang replacement. These findings motivate controlling the intervention and evaluating it across representations.

Evaluation design also affects interpretation. Babanejad et al. [9] distinguish preprocessing during representation learning from preprocessing for downstream tasks. Kuswanto and Tanamal [10] use repeated seeds and paired uncertainty estimates in Indonesian app-review comparisons, with a preprocessing ablation for emotion classification. Repeated runs and ablations therefore have clear precedents. Bestgen [11] emphasizes reporting differences with confidence intervals, supporting direct comparisons of normalization effects rather than conclusions based only on separate model scores.

Together, these studies motivate a controlled evaluation that distinguishes representation, preprocessing policy, and the point at which substitution is applied. The present study combines one fixed dictionary intervention, separate word, character, and contextual models, and shared audited splits on SmSA and PRDECT-ID [12]. Existing resources and consolidation efforts such as NusaCrowd [13] enable this assessment without new scraping or annotation. Its contribution is the joint comparison of separately trained pipelines and input changes to fixed models, with paired uncertainty and error transitions, within two specified sentiment settings.

III. Methodology

A. Datasets and Partitioning

PRDECT-ID contains 5,400 Indonesian Tokopedia product reviews across 29 categories, labeled positive or negative [12]. SmSA contains online comments and reviews with positive, negative, and neutral labels; its released train, validation, and test sets contain 11,000, 1,260, and 500 examples, respectively [6]. These are release sizes, not final sample counts. Only review text was used as input; ratings, emotion labels, and product metadata were excluded. SmSA's baseline uses its released, already preprocessed text without additional slang normalization.

The audit removed 112 canonical duplicates from PRDECT-ID and 81 from SmSA's training partition; no missing/invalid records or conflicting-original-label groups were found. Records sharing canonical original or normalized text were kept within one partition, forming the shared splits in stage 1 of Fig. 1. PRDECT-ID used approximately 70/15/15 group partitions, while SmSA retained its official assignments and unchanged validation/test membership. Table I gives retained counts; Supplementary Table S4 reports class support. No canonical original or normalized text crosses partitions. The deterministic audit and assignment rules are detailed in subsection F.

TABLE I
Audited Dataset Partitions

Dataset	Train	Validation	Test
PRDECT-ID	3,702	793	793
SmSA	10,919	1,260	500

Using the audited partitions described above, this study compares two versions of the same reviews: minimally cleaned text (O) and dictionary-normalized text (N). As shown in Fig. 1, word-based SVM, character-based SVM, and IndoBERT are trained separately for each version, with model selection based only on validation performance. The selected models are evaluated on the same held-out reviews in two ways: comparing the original and normalized training pipelines, and changing the input while keeping each model fixed. These comparisons assess normalization benefits and identify errors corrected or introduced by input substitution. Stages 1–5 correspond to subsections A–E; the following subsections explain how the two input conditions and evaluation branches are implemented.

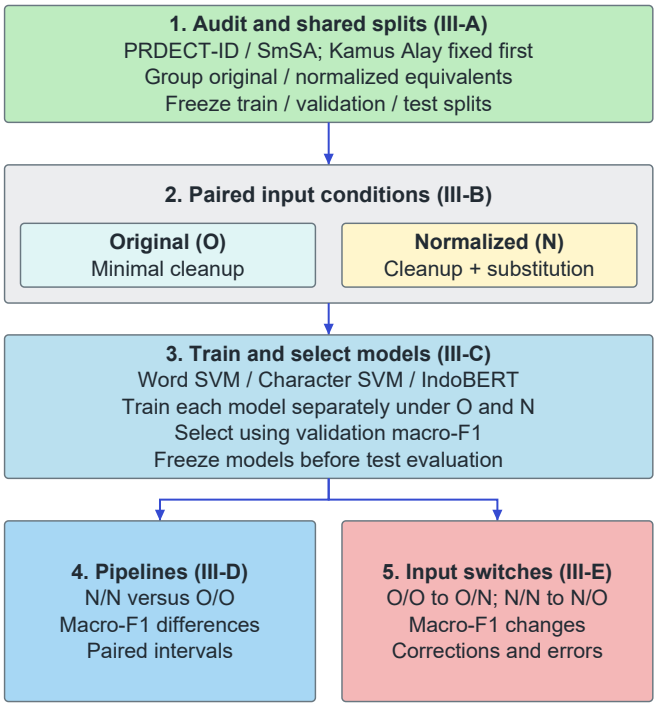


Fig. 1. Study workflow. A fixed dictionary informs the grouping audit before splits are frozen. The selected models feed two evaluation branches on the same test reviews: matched pipelines (stage 4) and fixed-model input switches (stage 5). In O/N notation, training condition precedes evaluation input.

B. Preprocessing and Dictionary Normalization

Stage 2 of Fig. 1 creates the paired input conditions. Let $O(x)$ denote minimal cleanup: Unicode NFC normalization

and trimming and collapsing whitespace. Let $N(x)$ apply dictionary substitution once to $O(x)$. Both conditions retain punctuation, emojis, and negation. Neither includes stemming, stopword removal, edit-distance correction, or other spelling transformations.

The normalizer uses a pinned version of Kamus Alay, associated with the Colloquial Indonesian Lexicon [1]. Dictionary keys and values undergo NFC normalization, case folding, and whitespace cleanup. Eligible keys are single tokens consisting of Unicode letters or decimal digits; replacement values may contain multiple words. Empty replacements and keys with multiple distinct formal alternatives are excluded. Identity mappings do not count as changes. Matching is case-insensitive on complete, maximal letter-or-digit tokens, leaving unmatched text and surrounding symbols unchanged. Replacements use the stored formal value and are not processed again within the same pass.

The pinned dictionary snapshot contains 15,006 rows. Restricting keys to Unicode letters and decimal digits yields 4,202 distinct eligible keys; 71 with conflicting replacements were skipped, leaving 4,131 active mappings. Replacement logs record tokens, replacements, and positions. Four normalized collision groups remain in PRDECT-ID and one in SmSA, none with conflicting labels. The policy was checked using constructed examples without manual inspection of test reviews and was fixed before evaluation.

C. Models and Training

In stage 3, the two datasets, three classifiers, and two preprocessing conditions formed 12 configurations; both conditions used the same retained examples and split assignments. The first two classifiers use word and character TF-IDF with a linear support vector machine (SVM). Word features include unigrams and bigrams over maximal letter-or-digit sequences and individual remaining non-whitespace symbols; character features span three through five characters, including spaces and punctuation. Both analyzers case-fold their inputs. Vocabulary and inverse document frequencies were fitted separately on training data under each condition, with minimum document frequency two, sublinear term frequency, smoothed inverse document frequency, and L_2 normalization. Limits were 50,000 word and 100,000 character features.

The contextual classifier fine-tuned the pretrained `indobenchmark/indobert-base-p1` checkpoint [6], with a linear head on its pooled representation and unweighted cross-entropy updating all layers. A serialized lowercase BERT WordPiece tokenizer uses the pinned vocabulary. Selecting the smallest length in $\{128, 256, 512\}$ covering at least 95% of training examples under each condition, including special tokens, yielded 128 for both datasets. Training truncation affects 14 original and 13 normalized PRDECT-ID reviews, and one review per condition in SmSA. A base checkpoint was chosen rather than a sentiment-fine-tuned model with unknown downstream training data.

Model selection used equal search budgets across preprocessing conditions: $C \in \{0.1, 1, 10\}$ for each unweighted L_2-penalized, squared-hinge SVM, and learning rates $\{2, 3, 5\} \times 10^{-5}$ for IndoBERT at tuning seed 42. Selection maximized validation macro-F1, evaluated after each of five transformer epochs. Validation ties favored smaller C, or earlier epochs followed by lower learning rates. Selected SVM C was 1 except for SmSA word-normalized ($C = 10$); selected IndoBERT learning rates were 2×10^{-5} except for SmSA normalized (3×10^{-5}).

Final SVM fits used seed 42, tolerance 10^{-4}, and a 20,000-iteration limit; convergence was checked before evaluation. IndoBERT was retrained from the base checkpoint with paired seeds 13, 37, and 73, selecting each checkpoint by validation macro-F1. AdamW used effective batch size 16 and the optimization settings in Supplementary Table S6; bias and layer-normalization parameters were exempt from weight decay, and pretrained dropout settings were retained. Supplementary Table S5 lists selected epochs. Validation data were not added to training, and test results did not guide configuration changes. Separate selection under each condition makes the primary effect a full tuning-and-preprocessing pipeline comparison.

D. Evaluation and Uncertainty

Stage 4 compares separately tuned pipelines: the original-trained model receives original text, and the normalized-trained model receives normalized text. This matched-pipeline comparison is distinct from the fixed-model branch in stage 5. Macro-F1 is the primary metric, computed over the fixed label set of each dataset:

$$M = \frac{1}{K} \sum_{k=1}^{K} \frac{2\mathrm{TP}_k}{2\mathrm{TP}_k + \mathrm{FP}_k + \mathrm{FN}_k}, \tag{1}$$

where K is two for PRDECT-ID and three for SmSA; a zero denominator contributes zero. Per-class F1 provides supplementary evidence. For each dataset and classifier, the main effect is

$$\Delta M = M(f_N(N(X)), Y) - M(f_O(O(X)), Y), \tag{2}$$

where f_O and f_N are the classifiers trained under the respective conditions, and (X, Y) is the shared test set. Transformer scores and paired differences are summarized by their mean and sample standard deviation across seeds. Absolute scores are not pooled across datasets.

Following the emphasis on effect sizes and uncertainty in NLP evaluation [11], 95% percentile bootstrap intervals accompany the differences. The analysis uses 2,000 paired bootstrap samples of test groups drawn with replacement and seed 42. The same samples are used across classifiers, conditions, and transformer seeds; each replicate averages the seed-specific transformer differences. Intervals describe test-sample uncertainty conditional on trained models; seed variability is reported separately.

To test whether normalization benefits the tested word model more than the alternatives, two contrasts are predefined per dataset:

$$D_{w,c} = \Delta M_{\text{word}} - \Delta M_{\text{char}}, \tag{3}$$

$$D_{w,b} = \Delta M_{\text{word}} - \overline{\Delta M}_{\text{IndoBERT}}. \tag{4}$$

Each bootstrap replicate recomputes these contrasts on identical sampled reviews. The six pipeline effects and four direct contrasts use nominal intervals, not simultaneous confidence guarantees. A positive effect for one model and an uncertain effect for another does not alone establish a difference between them.

Before classification experiments, 0.01 macro-F1 (one percentage point) was selected as a practical reporting reference. No further rationale was recorded; it is not a community standard. An interval wholly above 0.01 supports a practically meaningful gain within the tested setting; intervals crossing zero or the margin require caution. An interval containing zero does not establish equivalence, and this study is not a powered equivalence test. Per-class harms and introduced errors accompany aggregate interpretation.

E. Input Diagnostics and Error Analysis

Stage 5 reuses each selected classifier to predict both $O(X)$ and $N(X)$ without additional training or test-based tuning. Holding its parameters fixed isolates the effect of changing that model's input. These four combinations are denoted by O/O, O/N, N/O, and N/N, with the training condition first and evaluation input second. For each fixed model, prediction transitions count incorrect-to-correct, correct-to-incorrect, correct-to-correct, and incorrect-to-incorrect cases. These transitions complement the separately tuned pipeline comparison.

Changed and unchanged reviews are defined by $N(x) \neq O(x)$ before inspecting predictions. Identical inputs must yield identical predictions for a fixed model; this also serves as an implementation check. Coverage counts dictionary replacements and changed reviews; token and post hoc fixed-vocabulary vector diagnostics measure whether text changes reach the model input. Error analysis quantifies prediction-correctness transitions against the existing sentiment labels; it does not classify linguistic error types or add new annotations.

F. Reproducibility

Audit keys applied Unicode NFC normalization, case folding, and whitespace collapsing. Canonical-original duplicates with agreeing labels retained one record; conflicting-label groups would be excluded. SmSA priority was test, validation, then training, with source-order ties; PRDECT-ID used source order. Distinct originals colliding only after normalization were retained, including any differing labels.

Connected groups shared a canonical original or normalized string. PRDECT-ID groups were ordered by decreasing size, with SHA-256 of seed 42 and group identifier breaking ties. Greedy assignment to 70/15/15 targets minimized squared class-count deficits divided by target counts, plus a total-size term weighted by 0.25. SmSA retained official assignments; lower-priority members would be removed if a group crossed partitions. Dictionary, grouping code, tokenizer, and splits were jointly fingerprinted; changes require new overlap audits and, if affected, rebuilt splits before training.

The private `review_snapshot_v3` accompanies this manuscript with code, configurations, aggregate results, and file checksums. Its reviewer guide maps paper claims to artifacts; exact resource revisions and the protocol fingerprint also appear in the supplement. Review-level data, predictions, dictionary mappings, weights, private payloads, and credentials are excluded; reviewer access remains pending. The `REPRODUCIBILITY.md` guide documents retrieval, auditing, training, inference, analysis, and figure steps. Exact replay additionally requires privately retained selections, models, and predictions; fresh retraining need not reproduce the same weights. SVM training used Python 3.12.14 and scikit-learn 1.7.1; IndoBERT used Python 3.13.15, PyTorch 2.11.0, Transformers 5.16.1, and one Tesla T4. The cloud image and package inventory were recorded. Resource terms and release status are summarized in the supplement.

IV. Results

A. Matched-Pipeline Performance

Table II shows all four training/input cells and the matched-pipeline effects with intervals.

TABLE II
HELD-OUT MACRO-F1 AND MATCHED-PIPELINE EFFECTS

PRDECT-ID			
Train/ input	**Word SVM**	**Character SVM**	**IndoBERT**
O/O	0.9570	0.9595	0.9844
O/N	0.9583	0.9646	0.9836
N/O	0.9369	0.9495	0.9810
N/N	0.9646	0.9608	0.9810
Δ	+0.0076	+0.0013	−0.0034
95% CI	[−0.0013, +0.0177]	[−0.0076, +0.0101]	[−0.0097, +0.0025]
SmSA			
Train/ input	**Word SVM**	**Character SVM**	**IndoBERT**
O/O	0.7350	0.7214	0.8830
O/N	0.7366	0.7244	0.8836
N/O	0.7358	0.7342	0.9055
N/N	0.7414	0.7317	0.9032
Δ	+0.0064	+0.0103	+0.0202
95% CI	[−0.0135, +0.0297]	[−0.0046, +0.0267]	[+0.0037, +0.0381]

Training condition precedes evaluation input; $\Delta = N/N - O/O$. IndoBERT scores average three paired seeds. Intervals are nominal 95% paired group-bootstrap intervals conditional on fitted models.

On PRDECT-ID, the word SVM gained 0.0076 macro-F1, the character SVM gained 0.0013, and mean IndoBERT fell by 0.0034; all three nominal intervals include zero. On SmSA, the word and character SVM effects were 0.0064 and 0.0103, respectively, with intervals spanning zero. Mean IndoBERT improved by 0.0202, with an interval of [0.0037, 0.0381].

Although this interval is above zero, its lower bound is below the pre-experiment practical threshold of 0.01. Thus none of the six intervals supports a gain exceeding that threshold under the specified rule. Fig. 2 displays these estimates and intervals.

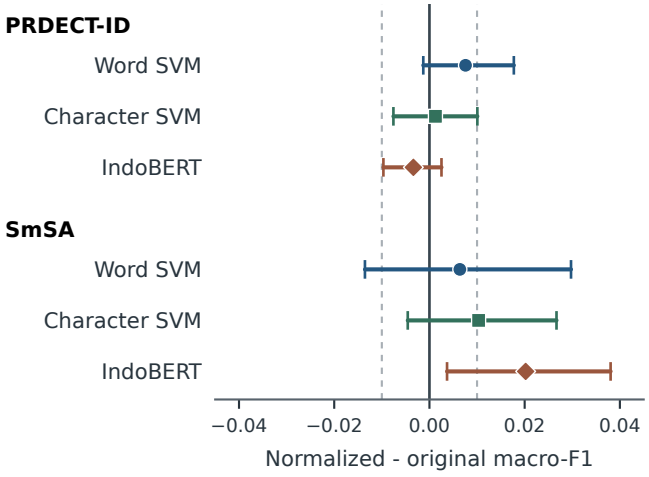


Fig. 2. Normalization effects by dataset and model. Points show normalized-minus-original pipeline macro-F1; bars show nominal 95% paired group-bootstrap intervals conditional on fitted models. Dashed lines mark the investigator-selected ± 0.01 reference margin.

The paired standard deviations of the three IndoBERT effects were 0.0090 for PRDECT-ID and 0.0088 for SmSA. Supplementary Table S1 gives the seed-specific scores and differences. These describe training variation separately from the conditional test-sample intervals defined in Section III.

B. Direct Representation Contrasts

Table III reports the four predefined differences between normalization effects. All four intervals include zero. In particular, the apparently larger SmSA IndoBERT effect does not by itself establish that its normalization effect differs from the word SVM's effect.

TABLE III
DIRECT CONTRASTS BETWEEN NORMALIZATION EFFECTS

Dataset	Contrast	Estimate	95% CI
PRDECT-ID	$D_{w,c}$	+0.0063	[−0.0038, +0.0178]
PRDECT-ID	$D_{w,b}$	+0.0110	[−0.0004, +0.0233]
SmSA	$D_{w,c}$	−0.0039	[−0.0303, +0.0226]
SmSA	$D_{w,b}$	−0.0138	[−0.0384, +0.0105]

$D_{w,c}$ and $D_{w,b}$ subtract character-SVM and IndoBERT effects, respectively, from the word-SVM effect. Intervals use the paired bootstrap defined in Section III.

C. Coverage, Class, and Fixed-Model Diagnostics

The normalizer made 1,527 replacements in PRDECT-ID and 121 in SmSA, changing 489 of 793 and 97 of 500 test reviews, respectively. Changed SmSA reviews comprised 52 negative, 30 positive, and 15 neutral examples. All text-changed reviews also had different word-token sequences and truncated BERT token IDs. Post hoc vector-change counts are reported in the supplement; some text changes left word TF-IDF vectors unchanged. These are operational matches and representation changes, not verified slang prevalence.

On SmSA, the word SVM's neutral-class F1 increased from 0.5263 to 0.5674, while negative and positive F1 declined from 0.8426 to 0.8337 and from 0.8363 to 0.8232, respectively. IndoBERT's mean neutral F1 increased from 0.7769 to 0.8248. The test set contains 88 neutral reviews, with limited normalization coverage as reported above. These class findings are descriptive, not separately adjusted tests; Supplementary Table S2 provides all class scores and supports.

The four cells in Table II also distinguish both fixed-model input directions. Applying normalization to an original-trained model gives $O/N - O/O$; supplying original inputs to a normalized-trained model gives $N/O - N/N$. On PRDECT-ID, the word SVM changed by $+0.0013$ in the first direction (10 corrected versus 9 introduced errors) and by -0.0277 in the reverse direction (7 corrected versus 29 introduced errors). Thus the input switch can both repair and damage predictions, and its direction matters. Supplementary Table S3 reports all directional differences, seed-mean transition counts, and unique-review counts.

For SmSA IndoBERT, the matched-pipeline change was $+0.0202$, whereas applying N to the original-trained model changed macro-F1 by $+0.0006$ and changed one prediction per seed on average. The reverse input switch on the normalized-trained model changed macro-F1 by $+0.0023$. These comparisons do not causally separate substitution, retraining, or tuning.

V. DISCUSSION AND LIMITATIONS

The direct contrasts leave RQ1 unresolved: the data do not establish greater normalization benefits for the word model or equivalence between representations. For RQ2, effects differ descriptively across datasets: SmSA IndoBERT has a positive nominal interval, whereas PRDECT-ID IndoBERT does not. The PRDECT-ID word interval permits a small negative effect or a gain above 0.01; SmSA IndoBERT permits positive gains below or above that margin. For RQ3, input switches both correct and introduce errors, and their effects depend on the direction of substitution.

The positive SmSA IndoBERT estimate is consistent in direction with Budianoor et al.'s normalization gain on culinary comments [4], but the uncertain PRDECT-ID result limits any broader recommendation. Bustamin et al.'s gains concern dictionary normalization combined with edit-distance correction [3], so they cannot be attributed to the same intervention tested here. Prichanti et al.'s representation-dependent preprocessing preferences [5] likewise motivate evaluating each configuration; the present direct contrasts do not resolve differences between representations. Different datasets, metrics, and preprocessing policies preclude direct comparisons of gain magnitudes.

For practice, the pipeline and fixed-model comparisons answer different decisions: whether to train with normalized text and whether to add substitution to an existing classifier. The

SmSA IndoBERT pipeline gain accompanied very small fixed-model changes, so the pipeline result alone does not justify inference-only normalization. Potential explanations involving feature sharing, subword robustness, or training adaptation remain post hoc hypotheses. Both decisions require evaluation on data representative of the intended use.

The findings are limited to two datasets, one dictionary policy, and one pretrained transformer. The small changed SmSA subset, particularly its 15 neutral reviews, limits class-specific interpretation. Dictionary matches are not independently verified slang annotations, and sentiment scores do not establish semantic preservation. IndoBERT uses only three seeds; the intervals exclude training uncertainty and offer no simultaneous coverage. Pretraining overlap cannot be excluded, and normalization-only runtime was not measured.

The SVM selection was bound to local checkpoint hashes before test inference. The frozen IndoBERT selection recorded versioned checkpoint identifiers but not whole-weight hashes; all 12 cloud files were hashed subsequently and reproduced their saved validation predictions before held-out inference. This supports execution provenance without a pre-freeze cryptographic binding of the transformer weight bytes.

VI. Conclusion

This study compared training with normalized text and switching inputs to fixed classifiers on shared audited reviews. The SmSA IndoBERT pipeline gain accompanied a much smaller inference-only change, while fixed-model switches both corrected and introduced errors. The contribution is evidence that a separately trained pipeline's benefit alone does not establish an inference-only benefit. Direct contrasts left representation differences unresolved, and no setting met the predefined practical-gain criterion. These findings support evaluating the two uses separately within each intended setting. Broader conclusions require additional datasets, dictionary policies, and measured preprocessing costs.

References

[1] N. A. Salsabila, Y. A. Winatmoko, A. A. Septiandri, and A. Jamal, "Colloquial Indonesian lexicon," in *2018 International Conference on Asian Language Processing (IALP)*, 2018, pp. 226–229.

[2] H. A. Wibowo *et al.*, "IndoCollex: A testbed for morphological transformation of Indonesian colloquial words," in *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, C. Zong, F. Xia, W. Li, and R. Navigli, Eds. Online: Association for Computational Linguistics, Aug. 2021, pp. 3170–3183. [Online]. Available: https://aclanthology.org/2021.findings-acl.280/

[3] A. Bustamin, A. A. Prayogi, D. Siswanto, M. Rafrin, and A. Nurdin, "Text normalization for Indonesian slang words in sentiment analysis development," *ICIC Express Letters, Part B: Applications*, vol. 16, no. 2, pp. 121–129, 2025.

[4] R. Budianoor, S. W. Saputro, F. Abadi, R. A. Nugroho, and A. Farmadi, "Quantifying the impact of text preprocessing on IndoBERT fine-tuning for Indonesian informal culinary sentiment analysis," *Journal of Computing Theories and Applications*, vol. 3, no. 4, pp. 564–581, 2026.

[5] L. A. Prichanti, K. R. Pangestutik, C. Suhaeni, and B. Sartono, "Preprocessing effects on Word2Vec and IndoRoBERTa for Shopee sentiment classification," *Journal of Applied Informatics and Computing*, vol. 10, no. 5, pp. 4660–4669, 2026.

[6] B. Wilie *et al.*, "IndoNLU: Benchmark and resources for evaluating Indonesian natural language understanding," in *Proceedings of the 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 10th International Joint Conference on Natural Language Processing*, K.-F. Wong, K. Knight, and H. Wu, Eds. Suzhou, China: Association for Computational Linguistics, Dec. 2020, pp. 843–857. [Online]. Available: https://aclanthology.org/2020.aacl-main.85/

[7] F. Koto, J. H. Lau, and T. Baldwin, "IndoBERTweet: A pretrained language model for Indonesian Twitter with effective domain-specific vocabulary initialization," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, M.-F. Moens, X. Huang, L. Specia, and S. W.-t. Yih, Eds. Online and Punta Cana, Dominican Republic: Association for Computational Linguistics, Nov. 2021, pp. 10 660–10 668. [Online]. Available: https://aclanthology.org/2021.emnlp-main.833/

[8] A. B. Siva, Robet, and L. Hoki, "Comparison of IndoBERT and SVM performance in sentiment analysis of digital education platforms," *Sinkron: Jurnal dan Penelitian Teknik Informatika*, vol. 10, no. 1, pp. 64–74, 2026.

[9] N. Babanejad, A. Agrawal, A. An, and M. Papagelis, "A comprehensive analysis of preprocessing for word representation learning in affective tasks," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, D. Jurafsky, J. Chai, N. Schluter, and J. Tetreault, Eds. Online: Association for Computational Linguistics, Jul. 2020, pp. 5799–5810. [Online]. Available: https://aclanthology.org/2020.acl-main.514/

[10] N. G. W. Kuswanto and R. Tanamal, "Benchmarking IndoBERT and IndoBERTweet on emotion and sentiment of indonesian app reviews," *Sistemasi: Jurnal Sistem Informasi*, vol. 15, no. 10, pp. 4260–4273, 2026.

[11] Y. Bestgen, "Please, don't forget the difference and the confidence interval when seeking for the state-of-the-art status," in *Proceedings of the Thirteenth Language Resources and Evaluation Conference*. Marseille, France: European Language Resources Association, Jun. 2022, pp. 5956–5962. [Online]. Available: https://aclanthology.org/2022.lrec-1.640/

[12] R. Sutoyo, S. Achmad, A. Chowanda, E. W. Andangsari, and S. M. Isa, "PRDECT-ID: Indonesian product reviews dataset for emotions classification tasks," *Data in Brief*, vol. 44, p. 108554, 2022.

[13] S. Cahyawijaya *et al.*, "NusaCrowd: Open source initiative for Indonesian NLP resources," in *Findings of the Association for Computational Linguistics: ACL 2023*, A. Rogers, J. Boyd-Graber, and N. Okazaki, Eds. Toronto, Canada: Association for Computational Linguistics, Jul. 2023, pp. 13 745–13 818. [Online]. Available: https://aclanthology.org/2023.findings-acl.868/